Why reading more helps or protects: the mechanism

Simulation 2, `task003-symmetric` and `task003-nosolution`

Step 2 of the key-phenomena study (Campaign A); 20 October 2026

Run. `results/simulation_2/2026-10-20_sim2_reading_mechanism`: both setups, 8 memories ρ, $q \in \{3, 6, 9, 12\}$, false controller (target A2) and silent arm, $q_c = 12$, $\lambda_c = 1$, gate on, $B = 30$, $W = 48$, 1,000 paired episodes, each agent's memory recorded every 0.25 and every board post (128 settings, 5.7 GB). The 96 settings at $q = 3, 6, 12$ reproduce the earlier runs exactly. G, D: as in step 1; all at $t = 30$. Bars: 95% intervals.

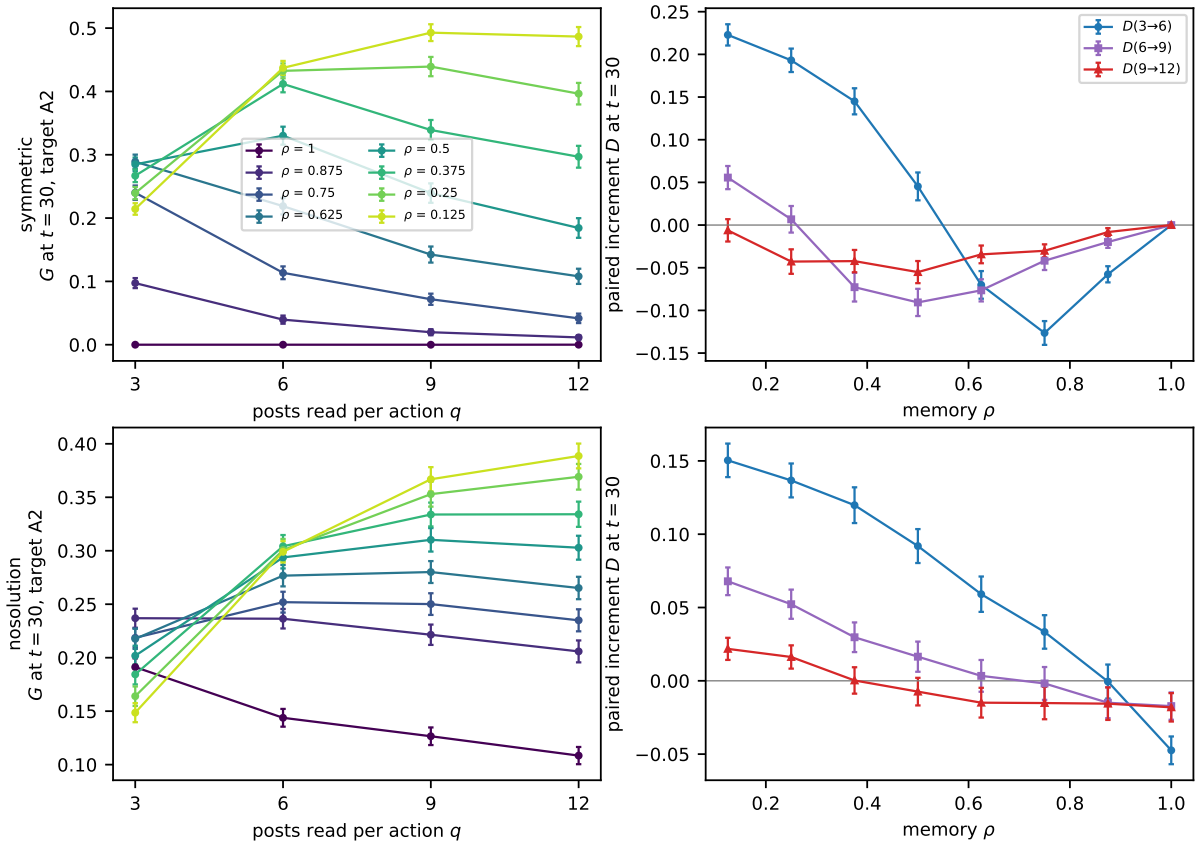


Figure 1: Left: G against q, one line per ρ. Right: $D(3\to6)$, $D(6\to9)$, $D(9\to12)$ against ρ. Top: symmetric; bottom: nosolution.

- The controller's best q falls as memory grows: symmetric 9 ($\rho \le 0.25$), 6 (0.375–0.5), 3 ($\rho \ge 0.625$); nosolution 12 ($\rho \le 0.375$), 9 (0.5–0.625), 6 (0.75), 3 ($\rho \ge 0.875$).
- Each further step of q changes G less when agents forget, and turns negative at lower ρ.

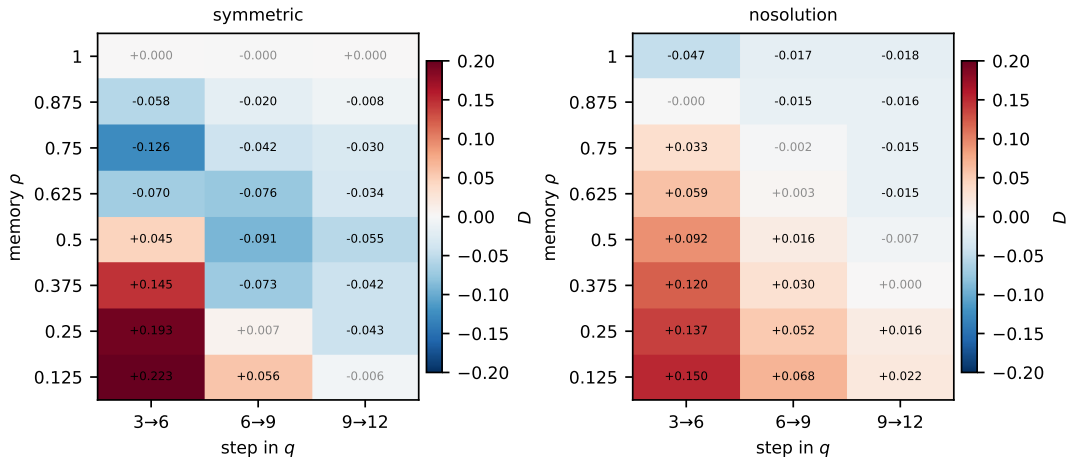


Figure 2: D for each step of q and each ρ (grey number: the 95% interval includes 0).

- Symmetric: D changes sign between $\rho = 0.5$ and 0.625 ($3 \to 6$), 0.25 and 0.375 ($6 \to 9$), 0.125 and 0.25 ($9 \to 12$).
- Nosolution: between 0.75 and 0.875 ($3 \to 6$), around 0.625–0.75 ($6 \to 9$), around 0.375–0.5 ($9 \to 12$).
- The crossover contour descends smoothly, by 0.125–0.25 in ρ per step of q, and lies about 0.3 higher in ρ without a proof. With the equidistant q grid it is a gradual slope, not a sharp edge.

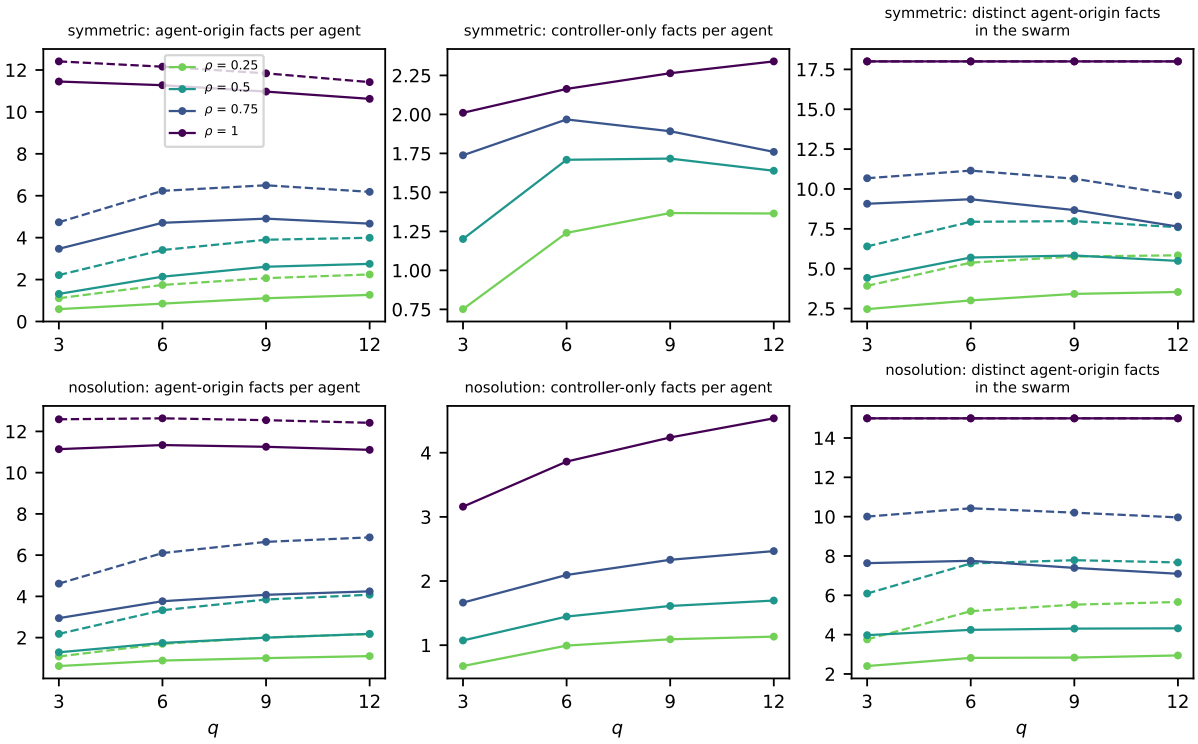


Figure 3: At $t = 30$, against q: agent-origin facts per agent, controller-only facts per agent, and distinct agent-origin facts held by anyone; false arm solid, silent dashed; $\rho = 0.25, 0.5, 0.75, 1$.

- Reading more lets agents hold more of their own evidence in both arms (false arm, $q = 3 \to 12$: up to $\times 2.2$ in symmetric and $\times 1.8$ in nosolution).
- At low ρ it also brings in more of the controller's facts (symmetric $\rho = 0.25$: $0.75 \to 1.36$ per agent); in symmetric at high ρ these level off or fall ($\rho = 0.75$: 1.74, 1.97, 1.89, 1.76).
- The controller lowers the swarm's coverage of agent-origin facts at every q and $\rho < 1$, by 0.9–2.4 (symmetric) and 1.1–3.5 (nosolution) distinct facts.

- Counts alone do not decide the sign: in nosolution at $\rho = 1$ agents hold more controller-only facts as q grows ($3.2 \to 4.5$) while G falls.

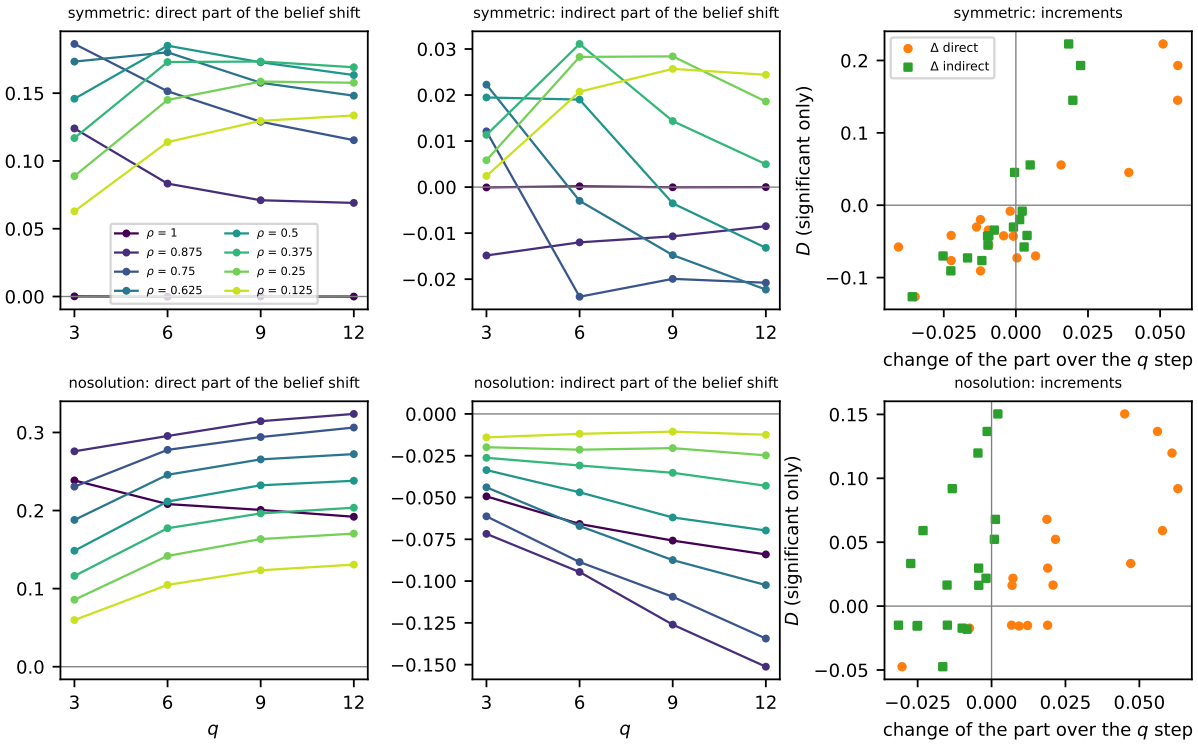


Figure 4: The false arm's extra belief in A2 over the silent arm, split exactly into a *direct* part (what the controller-only facts add: belief from all remembered facts minus belief from the agent-origin facts alone) and an *indirect* part (which agent-origin facts are kept: belief from them minus the silent arm's belief), against q, one line per ρ. Right: the change of each part over a q step against D (significant D only).

- The sign of D follows the change of the direct part in 17 of 19 significant increments (symmetric) and 15 of 19 (nosolution); correlations 0.89 and 0.83.
- Nosolution: reading more raises the direct part at every $\rho < 1$ (spreading), while the indirect part falls, most at high ρ (-0.015 to -0.031 per step at $\rho \geq 0.625$; integration of the agents' own evidence). D turns negative where the indirect fall outweighs the direct rise.
- Symmetric: at $\rho \geq 0.75$ the direct part falls with q at every step (-0.002 to -0.041): held alongside more agent facts, the controller's facts move beliefs less (dilution). At $\rho \leq 0.375$ both parts rise from $q = 3$ to 6.
- The two-function hypothesis is supported: reading spreads the controller's selection and integrates the agents' evidence; memory sets which wins.

Caveat. The split is on mean beliefs; G is on votes (drawn from beliefs). They agree in sign. One world per setup, $N = 24$.